

## ABSTRACT

In a multilingual world, navigating signboards written in unfamiliar languages poses significant challenges, especially for travelers and immigrants. To address this issue, this project introduces a solution titled "Multilingual Signboard Translator Using OCR", which integrates Optical Character Recognition (OCR), language translation, and speech synthesis into a unified system. The application allows users to either upload an image of a signboard or provide live audio input. Using Tesseract OCR, the system extracts text from images, and the live audio is converted into text using a speech recognizer. The extracted content is then translated into a user-selected language using the Google Translate API, and finally, the translated text is converted into speech via Google Text-to-Speech (gTTS). The application is built with the Flask web framework and features an intuitive interface, robust input clarity checks, and error-handling mechanisms to ensure a smooth user experience. This system serves as a practical tool for breaking language barriers in public spaces and can be particularly useful in fields like tourism, international transportation, and emergency services. The application has been tested for accuracy in OCR detection, translation quality, and responsiveness across different languages and formats, which provides us with more than 90% accuracy, and in case of a clearer image, it can be above 95%. Results indicate high reliability in extracting and translating text, even from images of varying clarity. By combining multiple technologies into a single, accessible web application, this project contributes to the growing need for inclusive communication tools in a globalized environment. Future work may include support for handwriting recognition, offline usage, and additional language dialects for wider applicability.

The outcome of this project is a functioning system that can accurately recognize and translate multilingual text from images of signboards, live audio in real time, and convert it into audio of the user's preferred language. This system helps overcome language barriers in public spaces by combining OCR (Optical Character Recognition) with translation technologies. The application helps to strengthen skills in AI/ML, OCR, and app development—applicable for the Ministry of Tourism, Government of India, travelling apps, freelance, or entrepreneurial paths. The application can be used prospectively for the. Real-time Images of signboards were collected across India. The SDG Indicators addressed by project are 4.4.1, 4.7.1, 9.5.1,9.5.2, 9.c.1,10.2.1, 10.3.1,11.2.1, 11.7.1, 11.7.2

### **Batch No: PWAD21A16**

Mohammad Rizwan (160721747007)

Keshetty Ramya Sri (160721747032)

Raparthi Bhoomika (160721747058)

### **Guide:**

Mrs. Unnati K,

Asst. Prof., Dept. of CSE